



A Deep Learning Method for Automatic Coronal Mass Ejection Feature Identification

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Abstract

Coronal mass ejections (CMEs), intense solar eruptive phenomena, are the primary drivers of extreme space weather storms on Earth. As space activities become increasingly frequent and infrastructure more reliant on space-based systems, the rapid and accurate detection and tracking of CMEs is critical. Here, we present a deep learning-based algorithm for automated CME feature extraction, comprising four key stages: image preprocessing, segmentation, tracking, and feature extraction. Initially, Solar and Heliospheric Observatory/Large Angle and Spectrometric Coronagraph C2 images undergo basic preprocessing. A segmentation model that integrates convolutional neural networks with transformer architectures is then employed to achieve pixel-level structural segmentation of CMEs. Subsequently, CME events in time series are tracked using Intersection over Union (IoU), Kalman filtering, and the Hungarian algorithm. Finally, critical physical parameters, including onset time, linear velocity, central position angle, and angular width, are extracted. Quantitative evaluations demonstrate that the proposed method performs well in metrics such as pixel accuracy, recall, F_1 score, and IoU. Qualitative analyses further reveal the capability of the algorithm to rapidly and precisely identify and track CMEs, including faint early-stage features. This approach is well suited for real-time CME warning and dynamic forecasting, offering robust data support for assessing the potential impact of CMEs on Earth.

Unified Astronomy Thesaurus concepts: [Solar coronal mass ejections \(310\)](#)

1. Introduction

Coronal mass ejections (CMEs) are among the most violent solar eruptive phenomena and serve as the primary drivers of extreme space weather storms on Earth. These events can trigger geomagnetic storms (J. Gosling et al. 1990; W. Fu et al. 2025), disrupt satellites and orbital space stations (F. Valach et al. 2009), elevate radiation exposure levels for astronauts during extravehicular activities (G. Reeves et al. 2003), and cause malfunctions in terrestrial electronic systems when impacting Earth (Z. M. K. Abda et al. 2020). As space-based activities become increasingly frequent and the reliance on space infrastructure grows, the ability to detect and track CMEs rapidly and accurately has become critical for mitigating potential hazards (T. Pulkkinen 2007).

Since the 1970s, coronagraph observations have facilitated the study of CMEs (O. Olmedo et al. 2008), enabling researchers to track their evolution from solar eruptions to interplanetary propagation. The launch of the Solar and Heliospheric Observatory (SOHO) spacecraft in 1995 marked a significant advancement in CME monitoring, primarily through the use of its Large Angle and Spectrometric Coronagraph (LASCO; G. Brueckner et al. 1995). In broadband white-light coronagraph images, CMEs typically appear as bright features moving radially outward from the Sun (O. Olmedo et al. 2008). Constructing comprehensive CME catalogs, which involves systematically listing and defining all detected CME events, has become an essential aspect of solar

and space weather research. These catalogs provide valuable data sets for understanding CME properties, such as their angular width (AW), central position angle (CPA), and kinematic parameters like velocity (J. P. Byrne 2015; R. Lin et al. 2024; J. Shan et al. 2024). Accurate identification and characterization of CMEs rely on advanced image processing techniques and algorithms to distinguish CME features from the surrounding coronal background. This growing body of work underscores the critical need for improved detection methods to support the study of CME dynamics and their space weather implications.

Existing CME catalogs can be broadly classified into manual and automated catalogs. One of the most well-known manual catalogs is maintained by the Coordinated Data Analysis Workshop (CDAW; S. Yashiro et al. 2004)⁵ Data Center. This catalog includes key parameters such as the first appearance time, CPA, AW, and linear velocity, and short video clips of CME events. These parameters are derived by researchers through the manual inspection of each coronagraph frame, a process that is both labor-intensive and time-consuming, particularly during periods of high solar activity when CMEs occur frequently. Moreover, the inherent subjectivity of manual visual measurements introduces potential biases. As the volume of solar observational data continues to grow exponentially, the need for automated CME detection methods has become increasingly urgent.

To address this demand, several automated CME detection techniques have been developed. E. Robbrecht & D. Berghmans (2004) introduced the Computer Aided CME Tracking



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⁵ https://cdaw.gsfc.nasa.gov/CME_list/

(CACTus; D. Berghmans et al. 2002)⁶ software package. This method leverages image processing techniques based on the Hough transform to identify CMEs. CACTus detects CMEs as bright ridges in two-dimensional time–height plots constructed from coronagraph images along specific position angles. CACTus has since established an online catalog for public use. O. Olmedo et al. (2008) developed the Solar Eruption Detection System (SEEDS),⁷ which employs a combination of morphological methods, region-growing algorithms, and thresholding techniques to track CMEs. SEEDS also maintains an accessible online catalog. Additionally, Y. Boursier et al. (2009) proposed an algorithm for Automated Recognition of Transient Events and Marseille Inventory from Synoptic maps,⁸ which was later updated to a next-generation algorithm by O. Floyd et al. (2013). This improved method uses adaptive filtering and segmentation techniques to identify CME structures based on synoptic map features. H. Morgan et al. (2012) and J. P. Byrne et al. (2012) introduced a novel deconvolution technique that separates coronagraph images into static and dynamic components. By isolating dynamic features, this method (CORIMP)⁹ identifies CME structures, which are subsequently characterized using multiscale edge detection algorithms and tracked over time-series data.

Recent machine learning approaches for CME detection have predominantly relied on texture, grayscale, and brightness features. However, due to the morphological diversity of CMEs, these handcrafted features and threshold-based techniques often fail to provide the desired level of generalization, resulting in suboptimal recognition accuracy across different events. In contrast, deep learning methods exhibit a superior capacity for feature extraction, enabling more robust and accurate identification of CMEs. For instance, P. Wang et al. (2019) introduced a convolutional neural network (CNN) to identify CMEs. Their approach involved training a LeNet model using labeled data from CME catalogs to classify CME images, followed by the use of deep feature transformation to locate common objects in image collections. Finally, image segmentation techniques were employed to detect CME structures. R. Alshehhi & P. R. Marpu (2021) utilized a pretrained VGG16 model to extract CME features and implemented unsupervised classification using K-means clustering based on principal component analysis (PCA) components. Morphological methods were subsequently applied to refine the results. R. Lin et al. (2024) adopted a CNN to classify LASCO images from SOHO into two categories: those containing CMEs and those without. They then used PCA and the Otsu (N. Otsu 1979) method to mark CME regions within the CNN-extracted feature maps. Building on this progress, J. Shan et al. (2024) employed a pretrained Visual Geometry Group model to generate binary labels indicating the presence or absence of CMEs, which were subsequently fed into a pretrained SegFormer model for precise segmentation. Additionally, polarization ratio (PR) techniques were applied to achieve three-dimensional reconstruction of CMEs.¹⁰

These advancements underscore the transformative potential of deep learning in CME research, offering more accurate and efficient solutions for CME detection, classification, and

characterization compared to traditional machine learning methods. While the aforementioned methods have demonstrated promising results, several limitations persist. Many of these approaches often yield coarse CME structures, particularly at the blurred boundaries, which complicates accurate localization and feature extraction of CMEs. Additionally, relying solely on the CDAW catalog can lead to erroneous CME detections, as certain CMEs may be either overlooked or misclassified. These inaccuracies can result in overestimation or underestimation of CME regions, ultimately producing CME parameters with significant errors.

To tackle these challenges, we introduce a novel deep learning framework that integrates CNNs with transformers for pixel-level segmentation of CME structures. Specifically, this hybrid architecture harnesses the strengths of both CNNs, which are proficient in extracting local features, and transformers, which excel at capturing global contextual information. To further refine segmentation accuracy, especially at CME boundaries, we incorporate an edge-awareness module during the feature extraction process. This module is tailor-made to capture intricate edge details, thereby enhancing the precision of CME boundary delineation. Subsequent to segmentation, we utilize a detection-based approach to track multiple CMEs over time. By amalgamating these cutting-edge techniques, our method provides a more holistic and precise solution for CME segmentation and tracking, addressing the limitations of current methodologies.

The structure of this paper is as follows. Section 2 provides a comprehensive explanation of our data sources and the preprocessing techniques employed. Section 3 introduces the proposed segmentation and tracking framework, detailing the feature extraction process. Section 4 presents the experimental results and analysis, showcasing the detection and tracking performance across several CME events and comparing the outcomes with those obtained using other classical methods. Finally, Section 5 discusses the implications of our findings and concludes the study.

2. Data and Preprocessing

Deep learning methods rely on large annotated data sets for network training, enabling the designed model to learn network parameters and subsequently make predictions on unseen data. In this study, the CME data set we constructed plays a crucial role in determining the final segmentation performance. Given the challenges of creating segmentation labels and the high cost of manual annotation, we aimed to achieve optimal segmentation performance with a minimal amount of labeled data. To this end, we manually annotated 100 images featuring a variety of CME morphologies. These 100 annotated images, which encompass cases ranging from narrow to wide (full halo), from near the solar disk to the edge of the field of view, from clear to blurred, from slow to fast, as well as instances of single CMEs and multiple coexisting CMEs, were collected between 2023 January 1 and 2023 December 30 and used as the training data set for the segmentation network.

For the test data set, we selected CME images captured between 2012 January 1 and December 31. Specifically, we annotated 52 images corresponding to three distinct CME events, along with the first 200 frames of CME imagery starting from 2012 January 1. This annotated data set was

⁶ <http://sidc.oma.be/cactus/>

⁷ <http://spaceweather.gmu.edu/seeds/>

⁸ <http://cesam.lam.fr/lascomission/ARTEMIS/>

⁹ <https://solarphysics.aber.ac.uk/Archives/CORIMP/>

¹⁰ <http://github.com/h1astro/CAMEL-II>

utilized to evaluate the model's performance in CME segmentation tasks.

In addition to this, we conducted a comprehensive 6 month evaluation using LASCO C2 coronagraph data, spanning from 2012 January 1 to June 30. During this period, approximately 18,000 images were analyzed, encompassing over 1000 CME events. This extensive assessment aimed to further validate the robustness and generalizability of our model across a diverse range of CME occurrences.

To develop an end-to-end deep learning method for automatic extraction of CME features, we sought to minimize input dependencies. This study employs white-light difference images from the LASCO C2 coronagraph on board the SOHO spacecraft¹¹ and additionally incorporates EUV observations. EUV data offer valuable information on CME source regions (Y. Li et al. 2024), which in turn improves the accuracy of predicting their arrival at Earth (H. Fu et al. 2021). The downloaded LASCO C2 coronagraph difference images, each with a resolution of 512×512 pixels, were screened to remove visibly distorted or anomalous images, reducing potential errors. To suppress noise in the raw images, a Gaussian blur (R. C. Gonzalez 2009) with a kernel size of 5×5 was applied to each image, following the method outlined by R. C. Gonzalez (2009). For precise annotation, we initially employed the labelme tool to delineate CME pixel regions, followed by fine-tuning with Photoshop, particularly at the CME boundaries, to ensure pixel-level accuracy.

Given the limited number of annotated images, data augmentation techniques were applied to both the original and labeled images to expand the data set. These included random horizontal flipping, rotation, adjustments to brightness and contrast, and gamma correction to modify overall image intensity. This process augmented the data set 20-fold, resulting in 2000 annotated training images (as illustrated in Figure 1).

3. Method

We present an automated method for extracting CME features, which requires only a sequence of CME differential images as input. The method autonomously identifies and tracks CMEs, extracting key parameters for each event, including the onset time, CPA, AW, and linear velocity. The overall framework, depicted in Figure 2, consists of three main components: segmentation, tracking, and feature extraction.

3.1. Recognition

The key to effective tracking and feature extraction lies in accurately segmenting CME structures. In this section, our goal is to achieve pixel-level segmentation of CME regions. Building on our previous work (P. Yang et al. 2023), which successfully integrated deep learning methods into solar image processing with advanced and satisfactory results, we make significant improvements to adapt the method for CME tasks under constrained time and low-resource deployment requirements.

Specifically, we streamlined the transformer architecture into a single-branch structure, reducing the number of headers and layers to enhance efficiency. At the same time, we retained the edge-aware module (EAM) to extract fine edge features of

CMEs. To ensure segmentation precision, we combined the convolutional block attention module (CBAM) and atrous spatial pyramid pooling (ASPP). The overall framework of our segmentation network is illustrated in Figure 3.

3.1.1. CNN-Transformer Hybrid Encoder for Feature Extraction

In the framework illustrated in Figure 3, the left side represents the encoder component. Image feature maps extracted by the CNN serve as inputs to the transformer. Patch embedding (A. Dosovitskiy 2020) is applied to the feature maps, where the input image has dimensions $H \times W \times C$, representing spatial resolution $H \times W$ and C channels. LASCO white-light difference images contain only a single intensity channel. To match the red, green, and blue input format expected by standard CNNs, we replicate this intensity data into the red (R), green (G), and blue (B) channels, so all three channels are identical. Although this adds no new information, it improves numerical stability and speeds up convergence during training. The image is divided into $P \times P$ -sized patches, resulting in $N = H \times W / P^2$ patches, where N denotes the number of patches and the input sequence length. Each patch, with dimensions $P \times P \times C$, is flattened into a vector. These N patch vectors are then concatenated to form a two-dimensional matrix of size $N \times (P \times P \times C)$, effectively serializing the image. Each patch X_p is projected into a D -dimensional embedding space through linear projection, as expressed in Equation (1). This embedding serves as the input representation for subsequent transformer layers,

$$z_0 = [x_p^1 E; x_p^2 E; \dots; x_p^N E] + E_{\text{pos}}, \quad (1)$$

where E represents the patch embedding projection, and E_{pos} denotes the position embedding.

The processed feature map is fed into the transformer layers, where feature extraction is performed using the multihead self-attention (MSA; A. Vaswani 2017) mechanism. The transformer encoder consists of L layers of MSA and multilayer perceptron (MLP) blocks, as described below:

$$Z_\ell' = \text{MSA}(\text{LN}(z_{\ell-1})) + z_{\ell-1}, \quad \ell = 1 \dots L, \quad (2)$$

$$Z_\ell = \text{MLP}(\text{LN}(Z_\ell')) + Z_\ell', \quad \ell = 1 \dots L, \quad (3)$$

where $\text{LN}(\cdot)$ denotes LayerNorm, followed by the application of residual connections at each block, and Z represents the image feature encoding.

In the encoder component, we also incorporate an EAM to enhance the segmentation of CME boundaries. The hybrid transformer encoder network we designed, while highly effective, is computationally intensive due to its large structure, resulting in slower inference speeds. After extensive experimentation, we optimized the model by reducing the number of layers and heads in the transformer backbone to one per layer, at the cost of a slight reduction in accuracy. This modification led to a significant improvement in inference speed.

3.1.2. Decoder

In the previous section, we achieved faster processing speeds by decreasing the number of layers and heads within the transformer architecture. To counteract any potential decline in accuracy and even bolster performance, we integrated the CBAM_ASPP module into the decoder component positioned on the right side of Figure 3. The

¹¹ <https://cdaw.gsfc.nasa.gov/images/soho/lasco/>

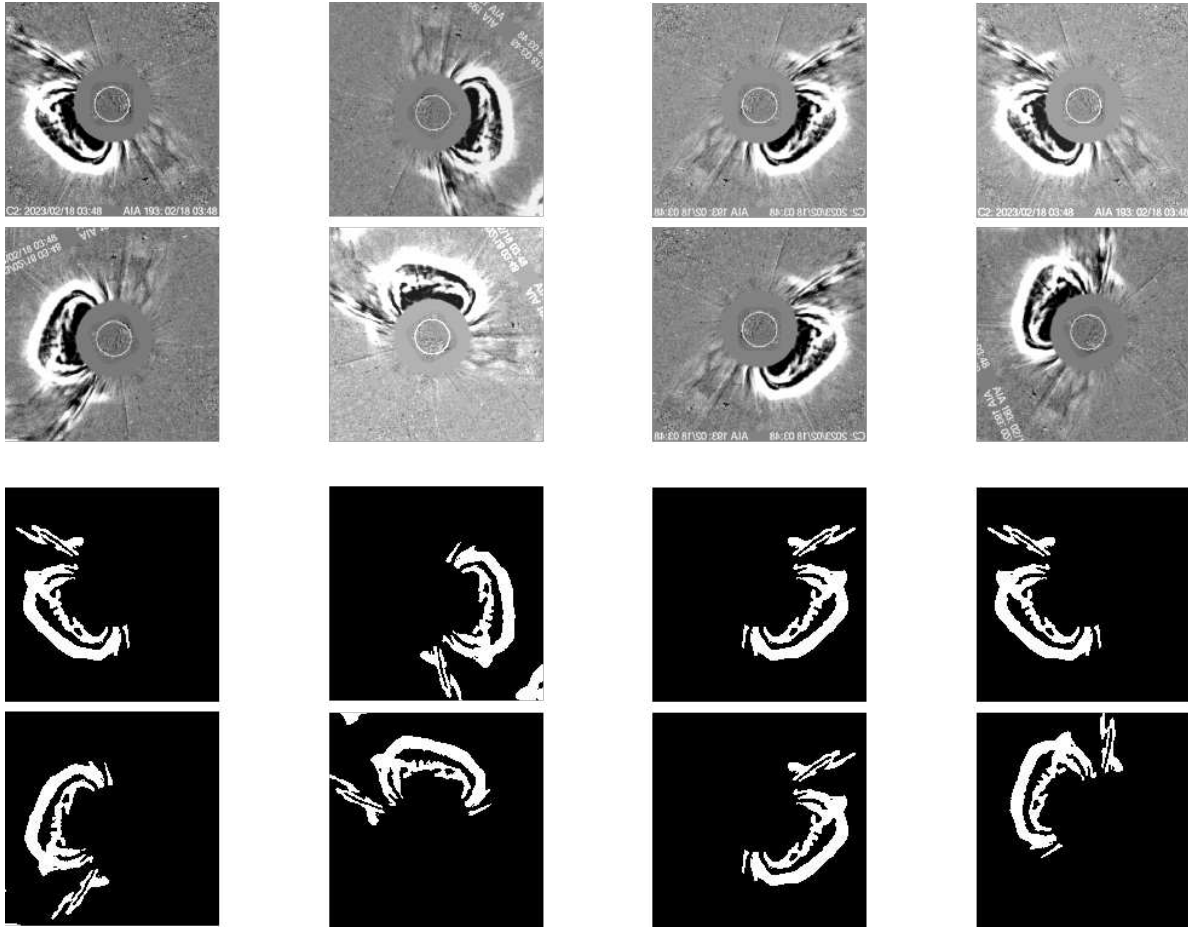


Figure 1. Expansion of the data set for both original and annotated images. The first image in the first row is the original image, and the first image in the third row is the image we annotated. The rest are our data-augmented images.

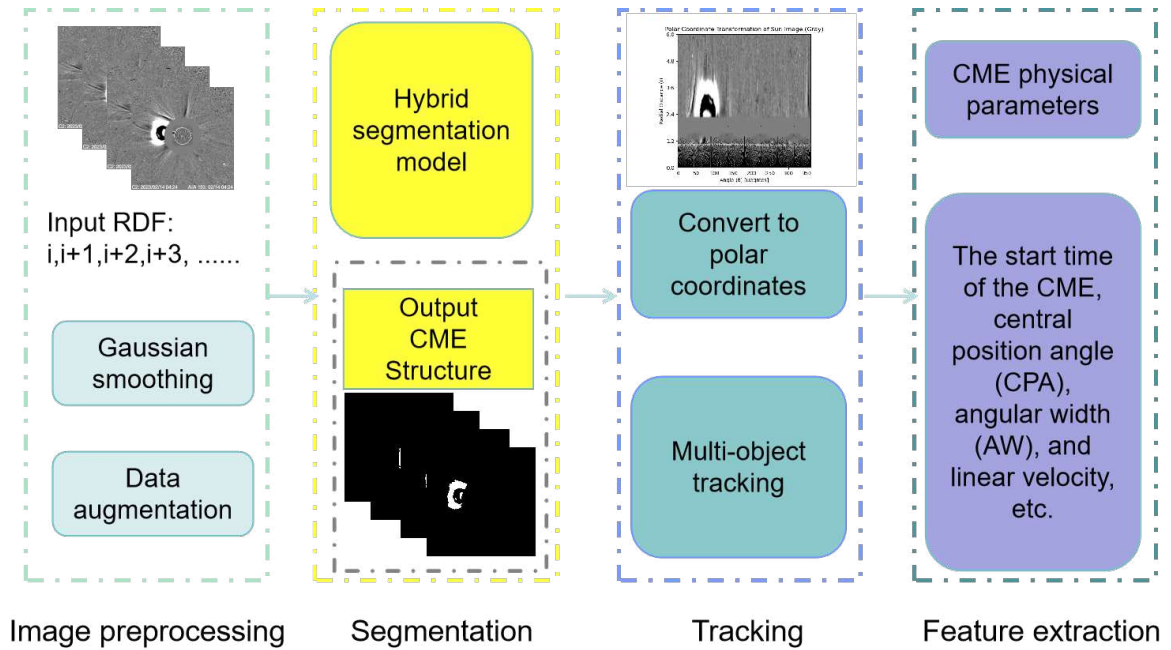


Figure 2. The proposed overall process for automatic identification of CME characteristics.

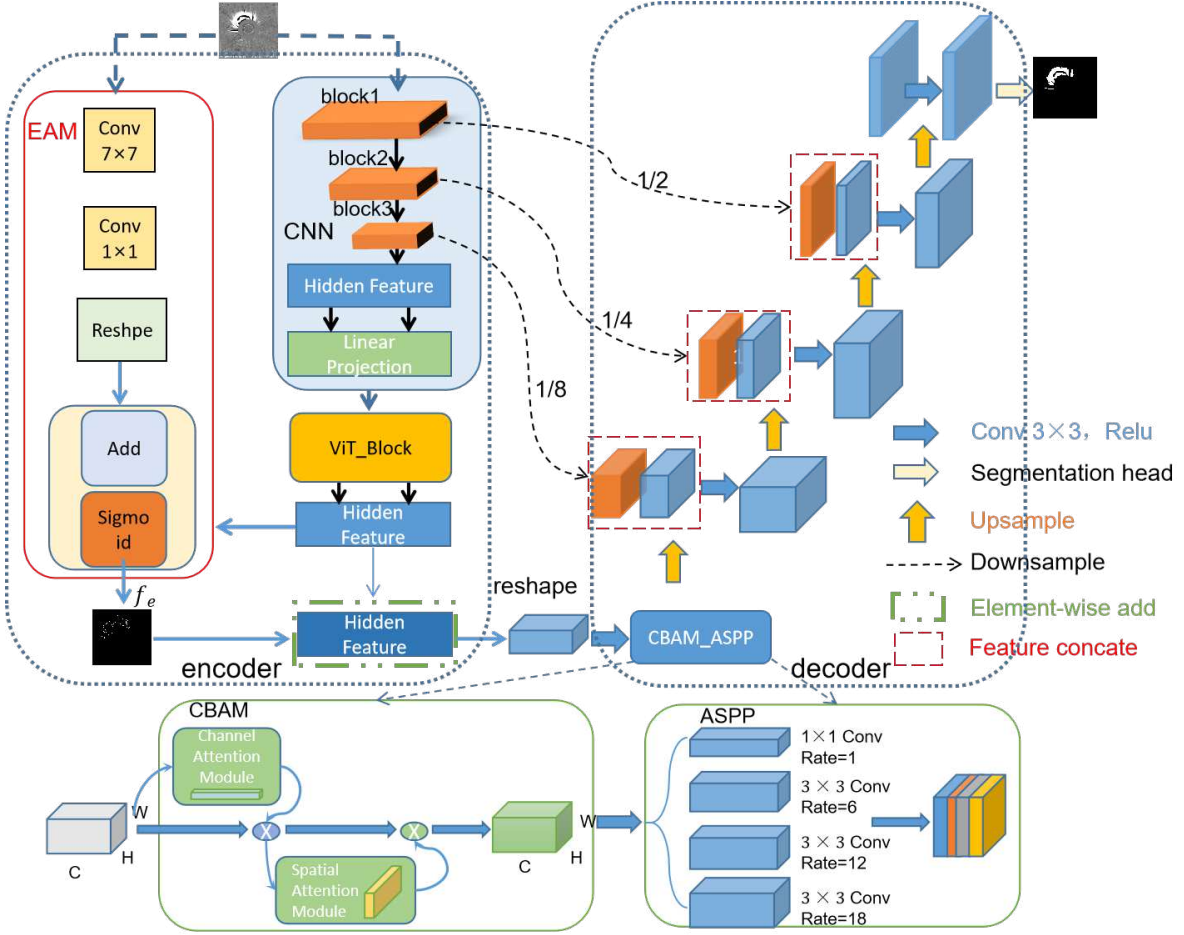


Figure 3. CME segmentation network (TransCME). The left half of the TransCME diagram, represented by the dashed box on the left, shows the encoder structure, while the right half, represented by the dashed box on the right, shows the decoder structure. The interpretation of CBAM-ASPP at the bottom shows the structure of the CBAM and ASPP parts in the decoder.

CBAM_ASPP consists of two parts, and its detailed construction process is outlined below.

After feature extraction by the encoder, we incorporate the CBAM (S. Woo et al. 2018). The CBAM combines channel attention and spatial attention modules, which individually process the input feature maps to enhance their representation.

Specifically, for the input feature map F , $F \in R^{C \times H \times W}$, we utilize the channel attention module to generate a one-dimensional channel attention map M_c , $M_c \in R^{C \times 1 \times 1}$, and the spatial attention module to generate a two-dimensional spatial attention map M_s , $M_s \in R^{1 \times H \times W}$:

$$F' = M_c(F) \otimes F, \quad (4)$$

$$F'' = M_s(F') \otimes F'. \quad (5)$$

The CBAM operates on the input feature map by passing it through two parallel layers: MaxPool and AvgPool. This transforms the feature map from dimensions $C \times H \times W$ to $C \times 1 \times 1$. The transformed feature map then undergoes the Share MLP module, which compresses the number of channels by a reduction factor of $1/r$ (where r is the reduction rate). The compressed channels are subsequently expanded back to the original number of channels, and rectified linear unit activation is applied to obtain two activated outputs. These outputs are element-wise added together and passed through a sigmoid activation function, resulting in the output of channel attention.

Finally, the channel attention output is multiplied element-wise with the original feature map to restore its dimensions to $C \times H \times W$:

$$M_c(F) = \sigma(MLP(\text{AvgPool}(F)) + MLP(\text{MaxPool}(F))) \\ = \sigma(W_1(W_0(F_{\text{avg}}^c)) + W_1(W_0(F_{\text{max}}^c))). \quad (6)$$

In the context of the aforementioned, the terms AvgPool and MaxPool refer to the operations of global average pooling and global max pooling, respectively. The acronym MLP stands for a shared multilayer perceptron module. Moreover, the symbol σ represents the sigmoid activation function. F_{avg}^c and F_{max}^c represent the average-pooled features and max-pooled features, respectively, $W_0 \in R^{C/r \times C}$, $W_1 \in R^{C \times C/r}$.

The output result of the channel attention is obtained by applying max pooling and average pooling to generate two $1 \times H \times W$ feature maps. These feature maps are then concatenated using the Concat operation. Subsequently, a 7×7 convolution is applied to transform the concatenated feature map into a single-channel feature map (empirical evidence suggests that a 7×7 kernel yields better results than a 3×3 kernel). Finally, the sigmoid function is applied to obtain the spatial attention feature map. The resulting feature map is multiplied element-wise with the original image,

restoring its size to $C \times H \times W$:

$$\begin{aligned} M_s(F) &= \sigma(f^{7 \times 7}([\text{AvgPool}(F); \text{MaxPool}(F)])) \\ &= \sigma(f^{7 \times 7}([F_{\text{avg}}^s; F_{\text{max}}^s])). \end{aligned} \quad (7)$$

The ASPP (L.-C. Chen et al. 2017) is a technique that samples the input features at different sampling rates, known as dilation rates, to extract features from multiple scales. These sampled features are then fused together to obtain the final feature representation.

The ASPP module consists of three main components: a 1×1 convolution, a pooling pyramid, and ASPP pooling. The pooling pyramid involves pooling operations with different dilation rates, allowing for flexible and customizable multi-scale feature extraction. Dilated convolutions, also known as atrous convolutions, differ from regular convolutions in their dilation rate, which controls the padding and dilation during the convolution operation. By adjusting the padding and dilation, different receptive field sizes can be achieved, enabling the extraction of multiscale information. The convolutional kernel size remains fixed at 3×3 throughout the process. The weight values generated by the CBAM are multiplied element-wise with the original input features, serving as the input for the ASPP module, resulting in the final feature representation.

In conclusion, the integration of the CBAM with the ASPP module offers a sophisticated and context-aware feature extraction process, enabling the model to focus on salient features and enhance its performance in object segmentation tasks.

3.2. Tracking

After identifying the CME structure, the next step is to track it within the time series. Prior to tracking, we remove isolated white connected regions with a pixel area of less than 100, as these are more likely to be dominated by noise. The image is then transformed from a Cartesian coordinate system (512×512) to polar coordinates (θ, R) (360×360), with θ representing the angular coordinate measured counterclockwise from the solar north pole, and R representing the radial distance from the occulting star. This transformation facilitates the subsequent tracking of CME attributes.

Considering that multiple CMEs may exist within the same image frame, it is essential to track and classify these distinct events in order to obtain fast and accurate CME features for subsequent analysis. The tracking method we propose, based on the combination of the Kalman filter and the Hungarian algorithm (H. W. Kuhn 1955), has also been widely applied in other domains for object tracking (B.-Z. Yang et al. 2024). This approach constitutes a fast and efficient solution for multiobject tracking. By leveraging our superior object detection method along with our tracking algorithm, we have significantly improved both the speed and accuracy of multiobject tracking. The tracking framework is illustrated in Figure 4.

To track the motion of CMEs across coronagraph image sequences, we adopt the Kalman filter framework, a recursive estimator widely used for dynamic object tracking. In this context, the system state x_k consists of the CME's position and velocity in the image plane at time step k , and it evolves

according to the following linear model:

$$x_k = Ax_{k-1} + Bu_{k-1} + w_{k-1}. \quad (8)$$

Observation equation:

$$z_k = Hx_k + v_k. \quad (9)$$

Time update (predicted):

$$\begin{aligned} \hat{x}_k^- &= A\hat{x}_{k-1} + Bu_{k-1}, \\ P_k^- &= AP_{k-1}A^T + Q. \end{aligned} \quad (10)$$

State update (updated):

$$\begin{aligned} K_k &= P_k^- H^T (HP_{k-1}H^T + R)^{-1}, \\ \hat{x}_k &= \hat{x}_k^- + K_k(z_k - H\hat{x}_k^-), \\ P_k &= (I - K_kH)P_k^-. \end{aligned} \quad (11)$$

Here, A is the state transition matrix, governing how the CME's position and velocity evolve over time. B is the control matrix, incorporating any external input u_k , which is often omitted in CME tracking due to the passive nature of motion. H is the observation matrix, mapping the current system state to the observable variables z_k , which in our case corresponds to the detected CME leading edge in each frame. w_k and v_k denote the process noise and measurement noise, respectively.

The Kalman filter also maintains a set of covariance matrices to model uncertainty. Q is the process noise covariance matrix, capturing uncertainty in the motion model (e.g., acceleration or CME expansion). R is the measurement noise covariance matrix, representing the uncertainty in the segmented CME front due to image noise or ill-defined boundaries. P is the error covariance matrix, describing the estimated uncertainty in the system state x_k itself.

The Kalman gain K is a key component computed at each step, optimally balancing the confidence between the predicted state and the new measurement. It ensures that the updated state estimate is influenced more by the measurement when the prediction is uncertain, and vice versa.

This formulation allows robust tracking of the CME's leading edge even under noisy, low-contrast conditions, by recursively estimating the most probable trajectory through time.

In other words, the Kalman filter updates the current state x_k by using the predicted state x_{k-1} from the previous time step and the observed state z_k from the current time step. The updated state x_k is then used as the predicted state x_{k-1} for the next iteration, with the process of prediction and update repeating iteratively. The Kalman filter operates in a continuous cycle of predict-update-predict-update.

In Figure 4, we introduce two key parameters: `max_age` and `min_hits`. `Max_age`, referred to as the maximum number of undetected frames, represents the threshold for consecutive frames in which a target remains undetected. Once this threshold is exceeded, tracking of the target will be terminated. Meanwhile, `min_hits` signifies the minimum number of detections required for a target to be considered valid. If the number of detections falls below this threshold, the tracking results for that target will not be reported. In this paper, `min_hits` is set to 0, and `max_age` is set to 3. In our tracking algorithm, the Hungarian algorithm constructs a similarity matrix based on the Intersection over Union (IoU) between the detection boxes (Dets) and the predicted tracking boxes (Trks). Each element of this matrix represents the similarity between a detection box and a tracking box, with the matching cost or

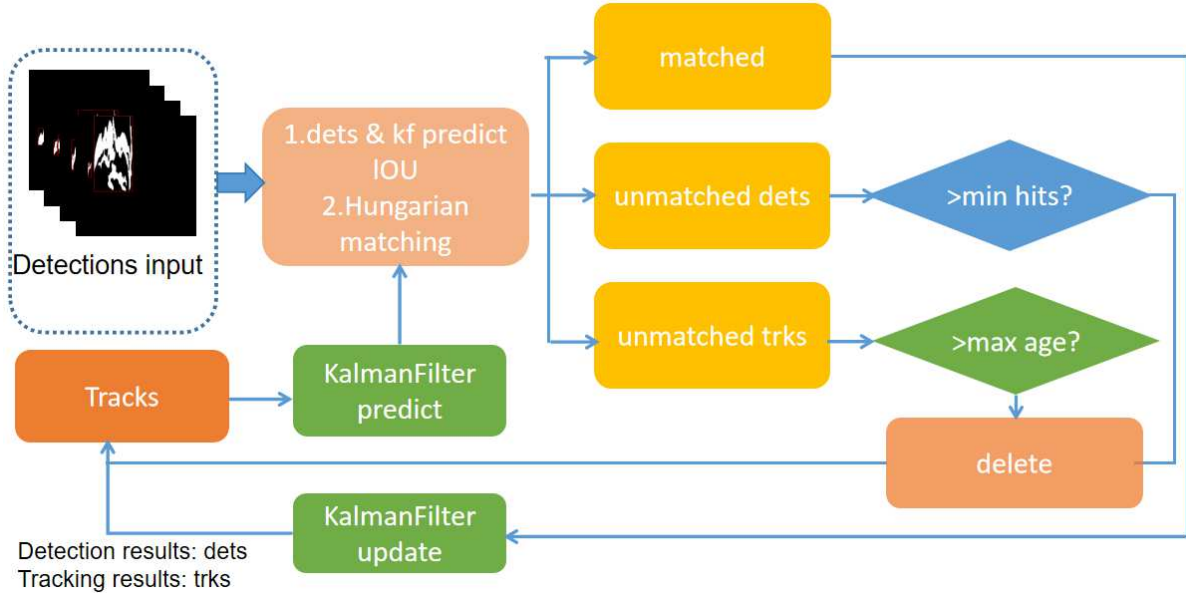


Figure 4. CME tracking flowchart. After detecting the CME structure, IoU, Kalman filter, and Hungarian algorithm are used to quickly track multiple identified CMEs. Max_age is the maximum number of frames lost in a single tracking of CME, while min_hits is the minimum number of frames successfully tracked for CME targets.

loss being calculated as $1 - \text{IoU}$. This matrix is then used to find the optimal set of matches that minimizes the matching cost, thereby ensuring the maximum IoU and, consequently, the minimal matching cost.

When objects enter or exit the frame, it is necessary to create or destroy unique identities accordingly. To initialize a tracker, we define that any detection with an IoU smaller than $\text{IoU}_{\min}$ indicates the presence of an untracked object. The tracker is initialized using the geometric properties of a bounding box with a velocity set to zero. Given that no velocity is observed at this point, the covariance of the velocity components is initialized with a large value to reflect this uncertainty. Furthermore, new trackers undergo a probationary period, during which the object must be associated with detections to accumulate sufficient evidence, thereby reducing the risk of false positive tracking.

3.3. Extract Feature

In the context of CME detection, CMEs are characterized as newly emerged, discrete, and bright white-light features exhibiting radial outward motion (O. Olmedo et al. 2008). To annotate CME regions within coronagraph images, we employed Adobe Photoshop for manual delineation, designating CME regions with a pixel intensity of 255 and non-CME regions with 0, thereby creating binary masks.

Afterwards, to refine the delineation of CME leading edges, we applied the Canny (J. Canny 1986) edge detection algorithm, which identifies regions of high-intensity gradients corresponding to CME boundaries. The Canny algorithm involves several steps: smoothing the image using a Gaussian filter to reduce noise, computing intensity gradients to detect edge strength and direction, applying nonmaximum suppression to thin out the edges, and utilizing double thresholding followed by edge tracking by hysteresis to accurately identify edge pixels.

During the training phase of our neural network, the edges detected by the Canny algorithm served as additional supervisory signals to guide the learning process. In the testing phase, the trained model was used to predict CME regions, and the leading edge pixels were extracted based on the model's output, ensuring precise identification of CME boundaries.

Upon completion of tracking, as illustrated in Figure 5, the initial position angle (PA) and final PA correspond to the left and right edge positions of the CME structure, respectively. The PCA is the average of the initial and final PA values, while the AW is the difference between the final and initial PA values. The height of the CME is defined as the distance from the bottom of the image to the top of the bounding box. As tracking progresses, the height of the CME over time is linearly fitted to derive its linear velocity. To obtain a representative linear velocity, the height is only fitted until the moment when the CME first reaches the image boundary.

4. Experimental Results

This section evaluates the effectiveness of our CME recognition and tracking method through a series of experiments. In Section 4.1, we present the evaluation metrics used to assess the segmentation results. In Section 4.2, we conduct a comparative analysis of our proposed segmentation network, which we call TransCME, with existing segmentation methods based on CNNs (B. Li et al. 2022) and transformers (E. Xie et al. 2021; J. Shan et al. 2024). Subsequently, in Sections 4.3, 4.4, 4.5, and 4.6, we assess the performance of our approach by comparing it with the results of CME events obtained from other established automatic tracking catalogs, including CACTus, CORIMP, and SEEDS, as well as the manual catalog CDAW. The results for these methods are sourced from their respective official websites, while the detection results for SegFormer (E. Xie et al. 2021; J. Shan et al. 2024) are derived from our reproduction of its methodology.

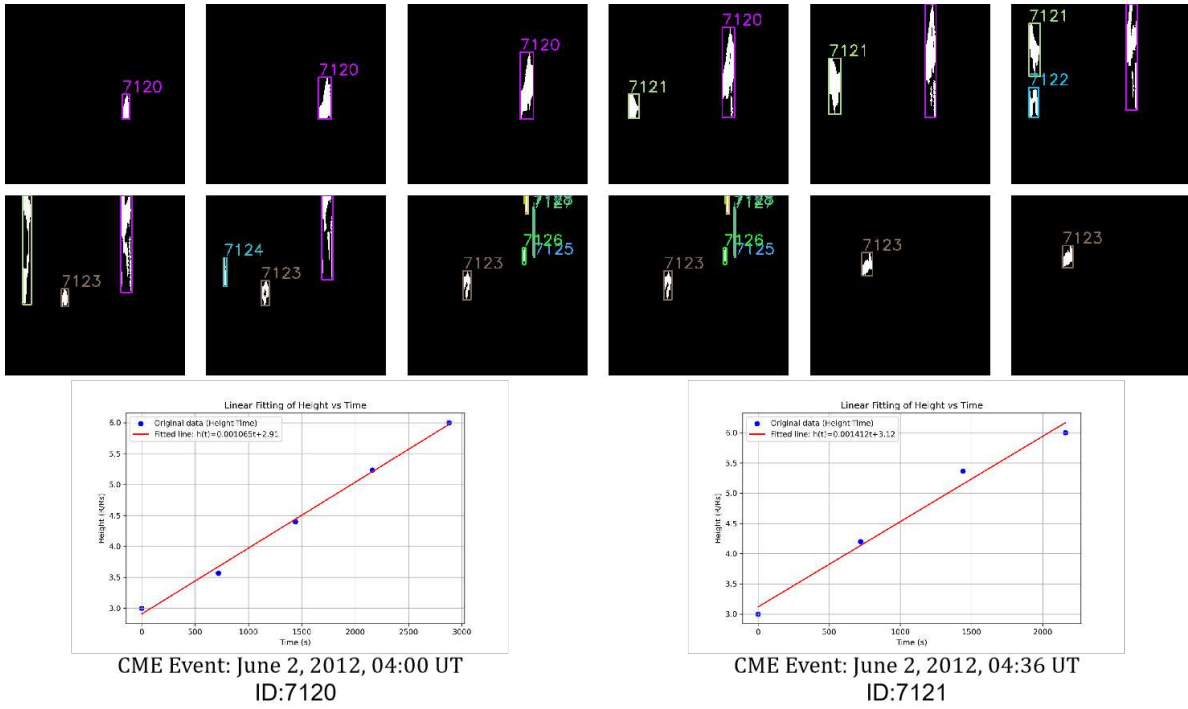


Figure 5. Tracking results and height–time fitting. The first two rows of images are time-series CME event tracking. ID 7120 was detected in the first frame, reached its highest point in the fifth frame, and continued to track until it disappeared. In the fourth frame, a new CME event appears and ID 7121 is detected until it disappears in the seventh frame. The bottom row of images shows the height over time fitting of these two CME events.

4.1. Evaluation Metrics

To evaluate the segmentation performance of our algorithm, we used four quantitative evaluation metrics for image segmentation (J. Long et al. 2015). These metrics include PA, recall, F_1 score, and IoU. In order to compute these metrics, we need the following values: true positives (TP), true negatives (TN), false positives (FP), and false negatives (FN), which are defined as follows.

TP represents the number of pixels predicted as positive that are actually positive. TN represents the number of pixels predicted as negative that are actually negative. FP represents the number of pixels predicted as positive that are actually negative. FN represents the number of pixels predicted as negative that are actually positive.

The PA is the proportion of pixels that are correctly classified among the total number of pixels. It is calculated using the following formula:

$$PA = \frac{(TP + TN)}{(TP + TN + FP + FN)}. \quad (12)$$

Recall is the proportion of true positive samples predicted among all the true positive samples. Generally, a higher recall indicates more positive samples correctly predicted by the model, indicating better performance:

$$\text{Recall} = \frac{TP}{(TP + FN)}. \quad (13)$$

The F_1 score is a combined metric of precision and recall, which considers both the accuracy and recall capabilities of the model. A higher F_1 score indicates a better balance between

accuracy and recall:

$$F_1 = \frac{2TP}{2TP + FP + FN}. \quad (14)$$

The IoU is the ratio of the intersection area between the predicted segmentation and the ground truth segmentation to the union area of both. It measures the segmentation accuracy of the model, where a higher value indicates a higher degree of overlap between the predicted and ground truth results:

$$\text{IoU} = \frac{TP}{(TP + FP + FN)}. \quad (15)$$

4.2. Segmentation Results

To demonstrate the effectiveness of our segmentation method, we compared TransCME with deep learning–based methods. For a fair comparison, we retrained CNN-based (DNet; B. Li et al. 2022) and transformer-based (SegFormer; J. Shan et al. 2024) models on the same training data set as TransCME. Experimental configurations adhered strictly to the parameters specified in the original papers. For SegFormer, we used the MiT-B1 encoder, Adam optimizer (D. P. Kingma & J. Ba 2014) with an initial learning rate of 0.0001, $T_0 = 30$, $T_{\text{mult}} = 2$, and a dynamic learning rate schedule. The loss function was cross entropy, and the random seed was set to 3407. In contrast, our model was trained using the SGD optimizer with a learning rate of 0.01, a momentum of 0.9, a weight decay of 1×10^{-4} , a batch size of 4, and a total of 5,000 training iterations.

Figure 6 illustrates the segmentation performance across several methods. While SegFormer (J. Shan et al. 2024) and our deep learning approach both performed well in segmenting

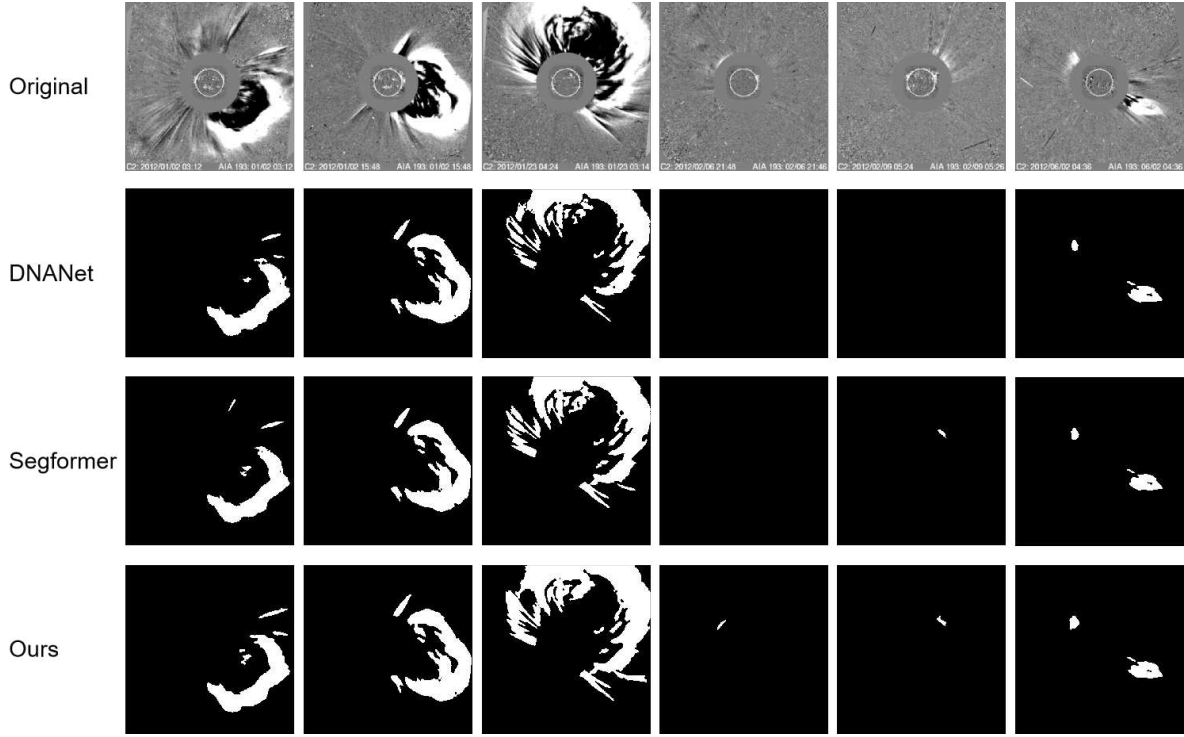


Figure 6. CME segmentation results. The first row represents some of the image frames we have selected, while the second, third, and fourth rows represent DNANet, SegFormer, and the detection results of our method, respectively.

Table 1
The Evaluation Metrics for DNANet, SegFormer, and TransCME

Data Time UT	DNANet				SegFormer				TransCME			
	PA	Recall	F_1	IoU	PA	Recall	F_1	IoU	PA	Recall	F_1	IoU
2012-02-06 21:48	0.998	0.75	0.85	0.74	0.999	0.91	0.93	0.86	0.999	0.96	0.97	0.94
2012-02-09 05:24	0.987	0.61	0.75	0.60	0.994	0.83	0.90	0.82	0.995	0.93	0.92	0.85
2012-03-23 19:27	0.993	0.69	0.81	0.68	0.996	0.86	0.91	0.83	0.998	0.95	0.95	0.90
2012-01-01 00:12 to 2012-01-02 19:12	0.996	0.72	0.82	0.70	0.997	0.79	0.87	0.77	0.998	0.95	0.91	0.83
Total	0.995	0.69	0.80	0.67	0.997	0.81	0.88	0.79	0.997	0.95	0.92	0.84

Note. For ease of comparison, we highlight in bold font the average highest indicator accuracy for each time span.

large and small CME structures, our method demonstrated superior integrity and structural accuracy. The first three columns in Figure 6 further emphasize the advantage of our approach, where the EAM significantly improved morphological accuracy, enabling more precise contour segmentation. Moreover, as shown in the last three columns of Figure 6, DNANet (B. Li et al. 2022) failed to detect faint CME structures, and the transformer-based method SegFormer was only occasionally able to detect weak CME structures. This is because TransCME leverages both the local detail sensitivity of CNNs and the global attention mechanism of transformers. Consequently, our approach is less affected by variations in brightness and morphology, enabling it to detect most faint and nascent CMEs while accurately extracting the CME onset time during tracking.

Quantitative results of the CNN-based method (DNANet), the transformer-based method (SegFormer), and our method (TransCME) are presented in Table 1. We conducted quantitative analyses on labeled CME events. The last row shows that DNANet achieved PA, recall, F_1 , and IoU scores of 0.995, 0.69, 0.80, and 0.67, respectively. SegFormer achieved

0.997, 0.81, 0.88, and 0.79 for the same metrics. TransCME achieved the best performance with scores of 0.997, 0.95, 0.92, and 0.84. The advantage of TransCME compared to methods utilizing CNNs is that CNNs struggle with balancing global and local information, leading to the exclusion of smaller CMEs situated near larger ones. While transformer-based methods captured most CME structures, their focus on global features at the expense of fine details resulted in slightly inferior performance on faint CME structures. Transformers' inherent reliance on global attention mechanisms often neglects image details at low resolutions, compromising pixel-level accuracy and producing coarse segmentation results.

Our integration of CNNs and transformers effectively addresses these limitations. To further enhance edge segmentation, we introduced the EAM, which improved edge detection accuracy. Additionally, the CBAM_ASPP module reduced training and inference times while enhancing model precision. When evaluated on real CME data sets, our method outperformed previous CNN and transformer-based

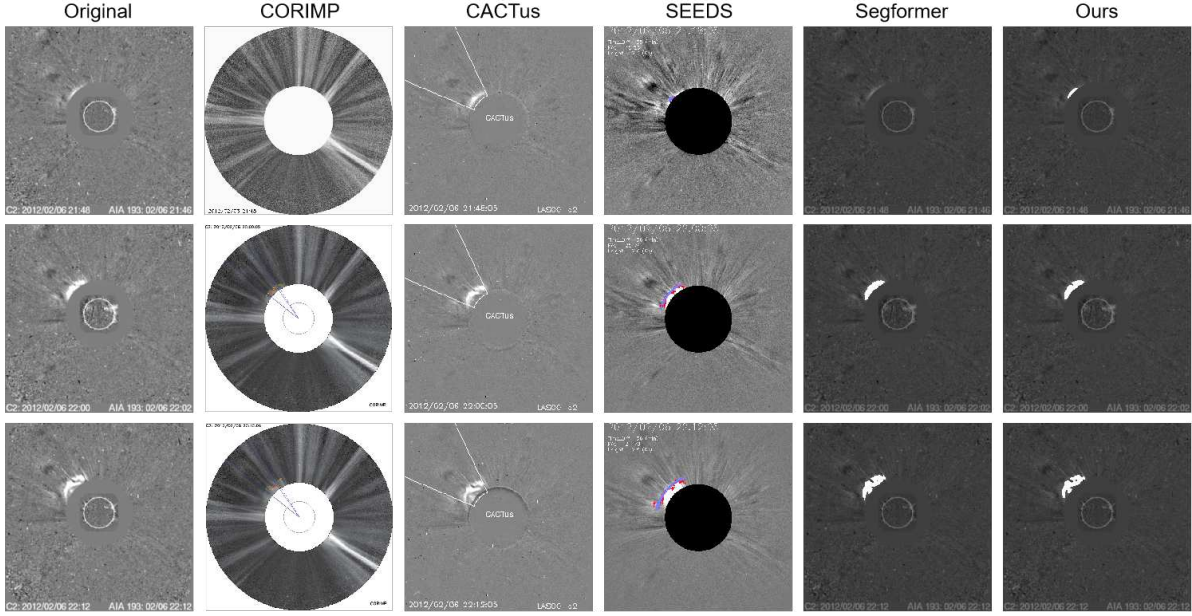


Figure 7. Comparison of our method with four classical methods for detection and tracking across consecutive time frames.

approaches, achieving superior segmentation accuracy and efficiency.

4.3. Six-month Comparative Analysis of CME Events

To assess the robustness of our CME detection methodology, we conducted a comprehensive analysis of LASCO C2 data spanning a 6 month period from 2012 January 1 to June 30. This data set comprised approximately 18,000 images, within which over 1000 CME events were identified.

For benchmarking purposes, we compared our detection results with entries from established CME catalogs, including the manually curated CDAW catalog and the automated CACTus and SEEDS databases. Our method detected 1392 CME events, surpassing the 990 events recorded by CDAW and the 1210 by CACTus, yet falling short of the 2124 events reported by SEEDS. The notably higher count in SEEDS aligns with findings from previous studies (O. Olmedo et al. 2008), suggesting a tendency to oversegment wide-angle CME events into multiple detections, potentially due to the misclassification of different fragments of a single CME as separate events.

To evaluate temporal correspondence, we cross-referenced the initial detection times of CMEs identified by our method with those listed in the CDAW, CACTus, and SEEDS catalogs, considering a temporal window of ± 1.5 hr. Our analysis revealed that 693 of our detected CMEs matched events in the CDAW catalog, accounting for approximately 70% of its entries. Preliminary assessments indicate that about 15% of the unmatched cases may result from discrepancies in event segmentation, where either our method or the CDAW catalog may have split or merged CME events differently. Furthermore, our detections corresponded to 929 events in the CACTus catalog (approximately 77%) and 1217 events in the SEEDS catalog (approximately 57%). The lower correspondence with SEEDS is consistent with its higher detection count, which may include multiple entries for single CME events.

Table 2

The Parameters Obtained from Our Catalog are Compared with Those from the CDAW, SEEDS, and CACTus Catalogs

Method	CPA (deg)	AW (deg)	Velocity (km s ⁻¹)
CDAW	180.8 $\pm$ 109.9	75.8 $\pm$ 78.6	449.7 $\pm$ 283.8
Ours	169.0 $\pm$ 106.5	27.1 $\pm$ 26.4	323.4 $\pm$ 254.7
SEEDS	180.1 $\pm$ 112.7	35.8 $\pm$ 39.8	343.3 $\pm$ 246.6
Ours	171.5 $\pm$ 113.1	21.9 $\pm$ 22.9	306.3 $\pm$ 268.7
CACTus	176.4 $\pm$ 107.7	36.1 $\pm$ 47.4	424.0 $\pm$ 295.6
Ours	175.8 $\pm$ 109.4	25.7 $\pm$ 25.0	311.3 $\pm$ 257.7

Beyond detection counts, we conducted a comparative analysis of key CME parameters—CPA, AW, and speed—between our catalog and those from CDAW, CACTus, and SEEDS. The mean values and standard deviations of these parameters are summarized in Table 2, providing insights into the consistency and discrepancies across different detection methodologies.

Table 2 clearly shows that the average AW and linear speed of CMEs obtained with our algorithm are markedly smaller than those listed in the CDAW, SEEDS, and CACTus catalogs. Two main factors account for this systematic bias. First, the definition of the CME front differs: we identify the front as the pixel with the maximum radial brightness gradient, whereas previous studies have shown that this position typically lags the “true” front (defined by a 2σ brightness threshold) by about 1–3 LASCO C2 pixels (≈ 0.02 – $0.07 R_{\odot}$), thereby underestimating the speed by roughly 20% and simultaneously narrowing the projected AW (E. Robbrecht & D. Berghmans 2004; J. P. Byrne et al. 2012). Second, the angular-width statistics differ: CDAW and CACTus report the instantaneous maximum AW during the event, whereas our study uses the time-averaged AW over the entire evolution. Because a CME is usually narrowest in its early stage, averaging further reduces the AW. The combination of these two mechanisms leads to systematically lower AWs and

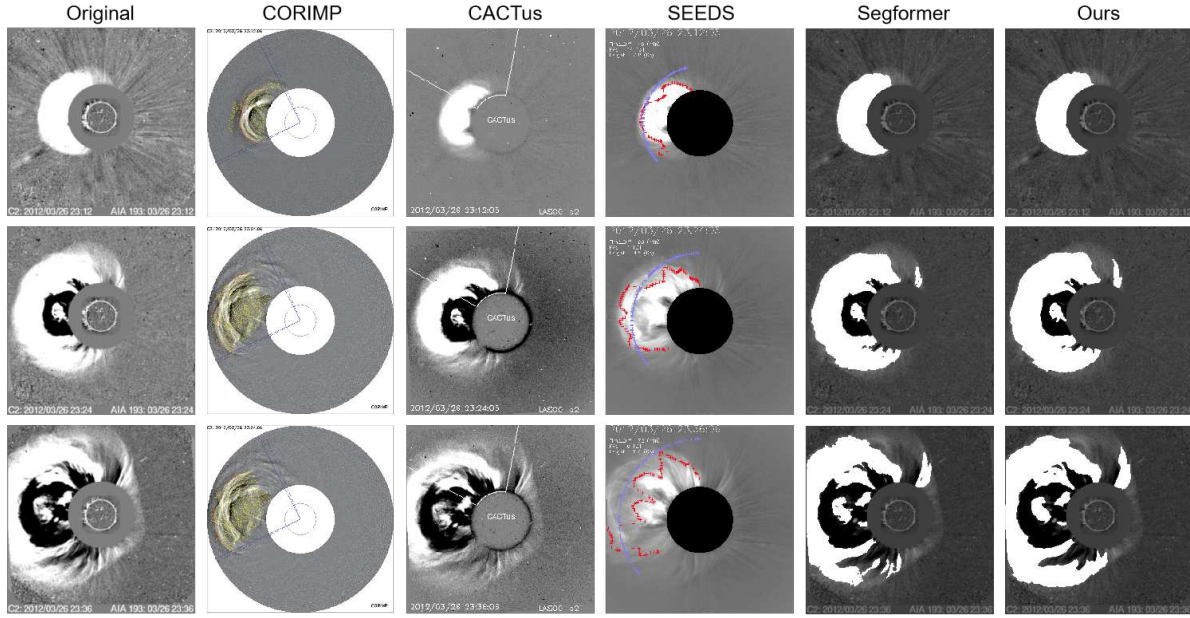


Figure 8. Comparison of our method with four classical methods for detection and tracking across consecutive time frames.

speeds in our results, fully consistent with the discrepancies observed in Table 2.

4.4. CME Event: 2012 February 6, 21:48 UT

To assess the versatility of our system, we extended our analysis beyond medium-speed and medium-scale CME events to include a diverse array of CME types. Specifically, we examined the system's performance in detecting broader CMEs, faster CMEs, and complex scenarios involving multiple overlapping CME events. This comprehensive evaluation demonstrates the system's capability to accurately identify and characterize CMEs across a wide spectrum of morphological and kinematic variations.

Figure 7 presents the original image captured at 21:48 UT on 2012 February 6, alongside the CME detection results from CORIMP, CACTus, SEEDS, SegFormer, and our proposed method. The first column displays three sequential LASCO C2 running-difference frame images with a temporal interval of 12 minutes. The second column shows the corresponding CME detection from the CORIMP method, where red points indicate the trajectory of the strongest and outermost front, and yellow points delineate the detected overall structure. The third column illustrates the CACTus detection, where the CME structure is enclosed within the angles formed by two straight lines. The fourth column provides the SEEDS detection results, with blue points marking the front edge and red points representing the approximate contour of the front edge derived through segmentation techniques. The fifth column showcases segmentation results using SegFormer, where white points correspond to the segmented CME structure, and gray areas represent unrelated background noise. Lastly, the final column displays the segmentation results from our proposed method, with white points indicating the segmented CME structure and gray regions denoting unrelated background. From the detection results in the first row, it is evident that our method achieves higher morphological accuracy compared to the other methods, particularly in delineating the CME contour with

Table 3
Key Parameters of the Event on 2012 February 6: Comparison of Our Method with Other Classical Methods

Method	Start Time	CPA (deg)	AW (deg)	Velocity (km s ⁻¹)
CDAW	21:48	48	69	274
CORIMP	22:00	44	17	218
CACTus	21:48	46	42	258
SEEDS	21:48	49	60	285
Ours	21:48	44	39	296

greater precision. This improvement is attributed to the effectiveness of our approach in segmenting the overall CME boundary.

Table 3 summarizes the key parameters of the CME event at 21:48 UT on 2012 February 6, as detected by different methods, including the manual CDAW catalog and the automated catalogs of CORIMP, CACTus, SEEDS, and our proposed method. The results from the aforementioned methods were obtained from their respective official sources. As shown in Table 3, most methods detected the onset of the CME at 21:48 UT, with the CPA ranging between 44° and 49°. In Figure 7, the AW determined by CORIMP is significantly underestimated at 17°, likely due to the overly narrow detection of the blue angle. By contrast, the AW obtained by CACTus, as suggested by the angle formed by the white lines, is more accurate. Our method estimates an AW of 39°, closely aligning with the CACTus result of 42°. Regarding the linear speed, our method calculates an approximate value of 296 km s⁻¹, which is consistent with the range of 218–285 km s⁻¹ reported by the other methods.

4.5. CME Event: 2012 March 26, 23:12 UT

Figure 8 illustrates the detection results for the CME event on 2012 March 26, at 23:12 UT, comparing five methods: CORIMP, CACTus, SEEDS, SegFormer, and our proposed approach. From top to bottom, the figure displays the detection

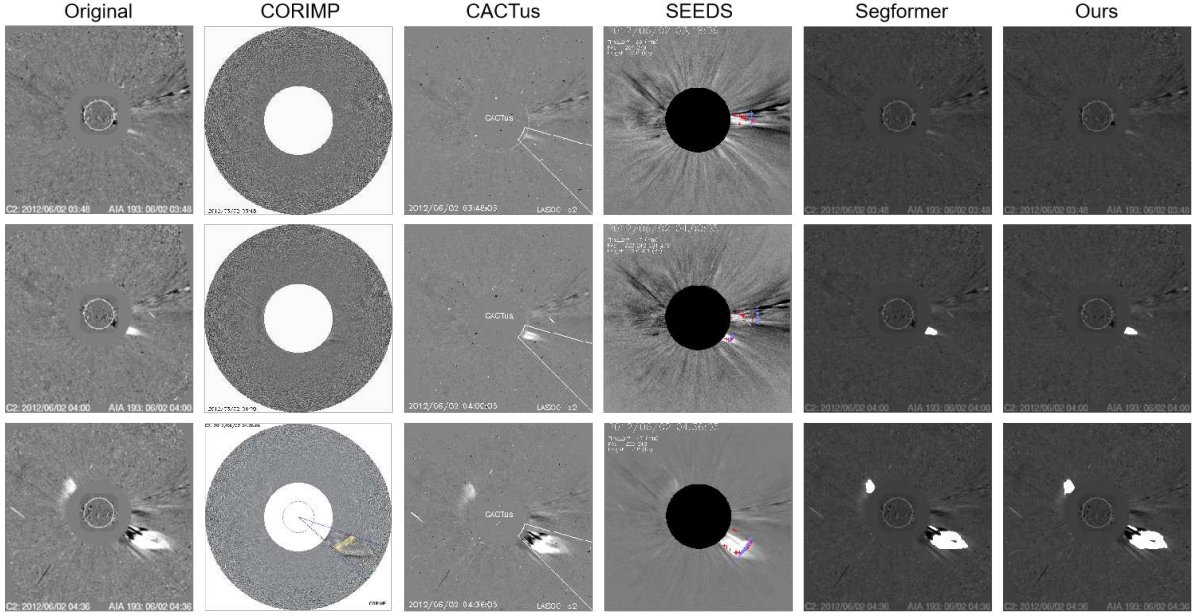


Figure 9. Comparison of our method with four classical methods for detection and tracking across consecutive time frames.

results for the three frames preceding the onset of the CME event. In this CME event, most detection methods, except for CACTus, successfully identified the approximate location of the CME. Compared to the SegFormer approach, our method delineated a more complete contour in the third frame, providing finer pixel-level segmentation. This enhanced precision facilitates more accurate extraction of critical parameters such as AW and velocity.

As illustrated in Table 4 and Figure 8, all methods detected the CME onset promptly. However, the CPA estimates varied significantly across the five methods, with our CPA measurement aligning most closely with the manually curated CDAW catalog. Additionally, our method yielded an AW larger than those reported by other automated catalogs, again closely matching CDAW’s values. Regarding velocity, this event qualifies as a fast CME. Both CDAW and CORIMP recorded speeds exceeding 1000 km s^{-1} , with our method estimating a velocity of 1644 km s^{-1} , closely mirroring CDAW’s data. In contrast, CACTus and SEEDS reported lower velocities of 976 km s^{-1} and 769 km s^{-1} , respectively.

4.6. CME Event: 2012 June 2, 03:48 UT

Figure 9 illustrates the detection results for a CME event that occurred on 2012 June 2 at 03:48 UT. The six columns represent the original image, followed by the detection outcomes from CORIMP, CACTus, SEEDS, SegFormer, and our proposed method.

In this CME event, multiple CMEs occurred simultaneously within single frames, presenting a complex scenario. In the initial frame at 03:48 (Figure 9, top row), only the CACTus method detected the CME’s presence. Notably, both our approach and the SegFormer method identified this CME; however, due to its incomplete structure comprising fewer than 100 pixels, both methods excluded it from this frame. The CORIMP and SEEDS methods failed to detect this CME at 03:48 but identified it in the subsequent frame at 04:00.

Table 4

Key Parameters of the Event on 2012 March 26: Comparison of Our Method with Other Classical Methods

Method	Start Time	CPA (deg)	AW (deg)	Velocity (km s^{-1})
CDAW	23:12	360	360	1390
CORIMP	23:12	71	88	1072
CACTus	23:12	25	72	976
SEEDS	23:12	65	131	769
Ours	23:12	88	167	1644

Table 5

Key Parameters of the Event on 2012 June 2: Comparison of Our Method with Other Classical Methods

Method	Start Time	CPA (deg)	AW (deg)	Velocity (km s^{-1})
CDAW	03:48	237	47	819
CORIMP	04:12	240	22	607
CACTus	03:48	240	30	656
SEEDS	04:00	234	29	629
Ours	04:00	239	23	741

Beyond the 03:48 CME event, a new CME was detected at 04:36 by our method, CDAW, and CACTus, whereas CORIMP and SEEDS did not identify this event. The CDAW catalog reported a CPA of 80° , an AW of 130° , and a velocity of 1175 km s^{-1} . CACTus recorded a CPA of 44° , an AW of 18° , and a velocity of 1106 km s^{-1} . Our method estimated a CPA of 47° , an AW of 22° , and a velocity of 983 km s^{-1} .

As shown in Table 5, the detected onset times for the 03:48 CME varied among methods, recorded at 03:48, 04:00, and 04:12, differing by approximately two frames. Regarding CPA, all catalogs reported values between 234° and 240° , indicating precise and consistent positional detection across methods. For AW, CORIMP reported the narrowest

measurement at 22° , while CDAW recorded the widest at 47° , with other methods' estimates falling within this range. In terms of velocity, our method estimated 741 km s^{-1} , closely aligning with CDAW's 819 km s^{-1} . In contrast, CORIMP, CACTus, and SEEDS reported velocities of 607 km s^{-1} , 656 km s^{-1} , and 629 km s^{-1} , respectively, showing greater deviations from the CDAW manual catalog.

5. Discussion and Conclusion

In this study, we present an advanced deep learning framework for the automatic segmentation and tracking of CMEs. Our method comprises four main stages: first, we apply noise reduction and data augmentation to the images. Next, a trained TransCME model is employed for precise pixel-level segmentation of the CME structures. Subsequently, we leverage IoU, Kalman filtering, and the Hungarian algorithm for efficient tracking of multiple CMEs identified by TransCME. Finally, physical parameters, including CME AW, CPA, linear velocity, and onset time, are extracted. Our approach integrates a hybrid architecture, combining the local feature extraction capabilities of CNNs with the global context awareness of transformer models. To enhance boundary accuracy, we incorporate an EAM, while the CBAM_ASPP module is used to improve segmentation precision and computational efficiency.

The creation of a specialized CME data set is crucial for training and evaluating our method. This training data set consists of 100 manually annotated training images with diverse morphologies. The test data set comprises 52 annotated samples from three distinct CME events, along with the first 200 frames of CME imagery starting from 2012 January 1, ensuring comprehensive coverage of morphological variations and a reliable assessment of segmentation performance. To enhance the scientific significance and relevance of this study, we further evaluated 6 months of CME events by comparing our results with those from the automated catalogs SEEDS and CACTus, as well as the manually curated CDAW catalog. Our method achieves PA, recall, F_1 score, and IoU of 0.997, 0.95, 0.92, and 0.84, respectively, on the test set, outperforming DNANet, which achieves PA, recall, F_1 , and IoU of 0.995, 0.69, 0.80, and 0.67, respectively, and SegFormer, which scores 0.997, 0.81, 0.88, and 0.79, respectively. Quantitative evaluation demonstrates that TransCME significantly outperforms existing CNN-based (e.g., DNANet) and transformer-based (e.g., SegFormer) methods in terms of key metrics such as PA, recall, F_1 score, and IoU.

By addressing the limitations of current approaches, such as the lack of global context in CNNs and the sensitivity of transformers to low-resolution details, TransCME offers more accurate and robust CME structure segmentation, even for faint and emerging features. Experimental results show that our method achieves superior segmentation and tracking accuracy for CMEs, particularly in complex or faint scenarios. Compared to classical methods such as CDAW, CORIMP, CACTus, SEEDS, and SegFormer, our approach not only excels in segmentation precision but also captures crucial morphological features for further analysis, such as AW and CPA.

In CME identification and tracking, the proposed CNN- and transformer-based method accurately detects faint and newly emerging CMEs, providing valuable data for subsequent predictions regarding the arrival of CME events on Earth

and their timing. As demonstrated by S. Yashiro et al. (2008), the CDAW catalog may miss many faint CMEs during solar activity maximum, a limitation that our method effectively addresses. Furthermore, in AW extraction, J. Burkepile et al. (2004) observed that the CDA catalog tends to overestimate AW values, with relatively small AW values possibly closer to the true value (R. Lin et al. 2024). The AW values extracted by our method are also smaller compared to those from the CDAW catalog. These findings underscore the potential of TransCME for real-world applications, enabling more precise temporal and spatial analysis of CMEs and contributing to the advancement of solar physics research.

In the future, we aim to build on this method to further explore the impact of CMEs on Earth. These CME feature parameters could serve as input for analyzing the effectiveness of CME events in space weather prediction and early warning systems.

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